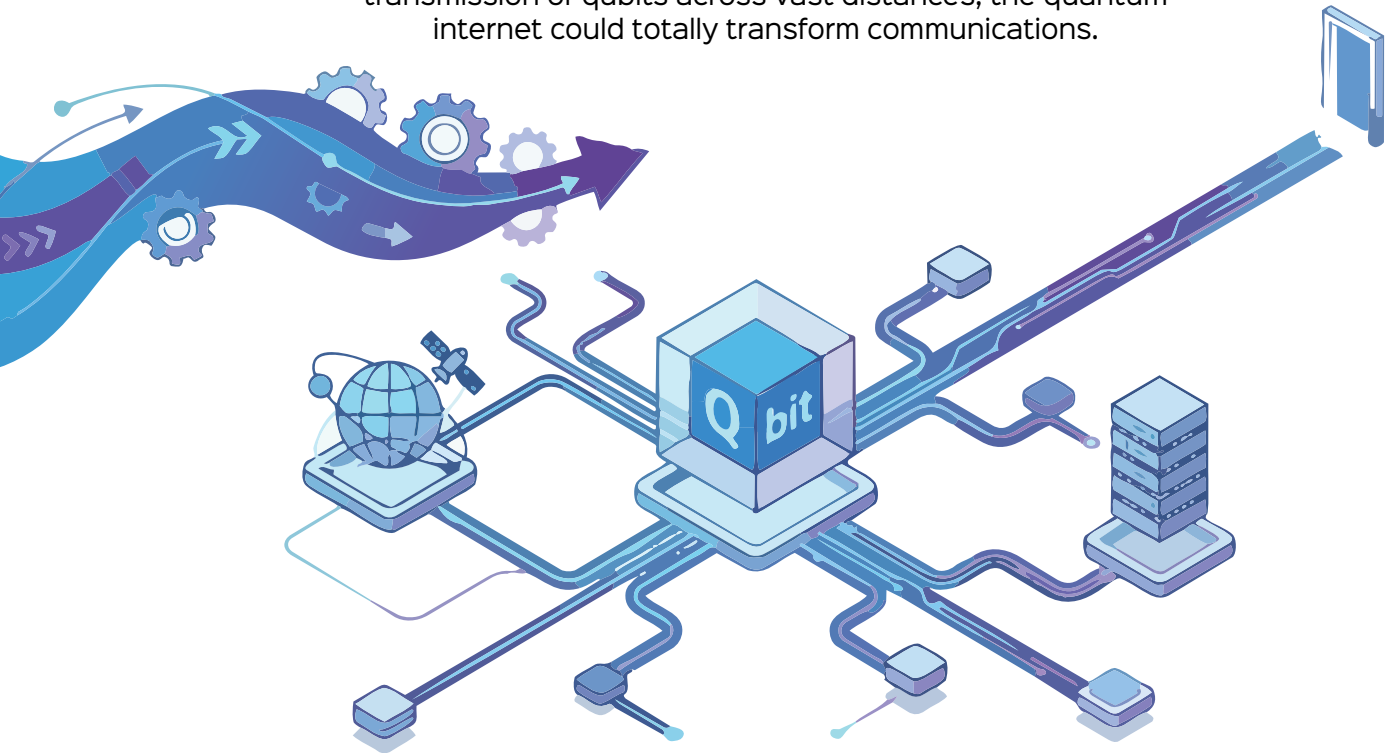


Quantum Networking and the Quantum Internet: The Road Ahead

Envisioned as a global network that enables the direct transmission of qubits across vast distances, the quantum internet could totally transform communications.



Quantum networking represents a radical shift in the way information is transmitted, processed, and secured. Unlike classical networks, which rely on the exchange of bits, quantum networks utilise the principles of quantum mechanics, particularly the properties of superposition and entanglement, to enable communication with unprecedented security and computational capabilities. The significance of quantum networking lies in its potential to revolutionise data transfer, bolster cybersecurity, and unlock new paradigms in distributed computing. As researchers and

technology professionals explore the possibilities of quantum communication, the prospect of a quantum internet emerges as a transformative milestone for the digital age.

From an engineering perspective, quantum networking is best viewed as a service for distributing non-classical correlations (entanglement) and, in limited cases, quantum states, using a tightly coupled quantum–classical control loop. Performance is constrained by channel loss, memory coherence time, detector noise, and synchronisation accuracy, which jointly determine achievable entanglement generation rate, fidelity, and end-to-end latency.

The vision: What is quantum internet?

The quantum internet is envisaged as a global network that leverages quantum entanglement and quantum states to transmit information securely and efficiently. A practical definition is that a quantum internet offers network-accessible primitives such as ‘entanglement as a service’ (deliver Bell pairs between nodes with specified fidelity and rate), quantum-secure key establishment, and time-synchronised quantum operations coordinated by a classical control plane. Consequently, capability must be described via metrics including